Contents

1. Design Philosophy

Goals, Not Parameters

Memcoin is built around goals, not parameters. Every parameter exists to serve a goal. When a parameter no longer serves its goal, the parameter changes. The goal does not.

The 8 GiB memory requirement exists so ordinary computers can compete fairly. If home hardware changes, the threshold changes with it. The 32 MB starting block size exists to guarantee zero-fee payments enter the next block. The block size adjusts automatically with demand up to a hard cap of 64 MB. The numbers are adjustable. The intentions are not.

Decentralisation as Consequence

Most networks lost this condition within years. Memcoin takes it as a design requirement.

Mining and validation carry identical resource requirements — 8 GiB and seconds per block. Anyone running a node can mine. The distinction between participant and validator does not exist on this network.

The Algorithm as a Guest

A mining algorithm makes demands on hardware. Most algorithms treat this as irrelevant — the machine exists to serve the computation, and wear, heat, and noise are acceptable costs.

Memcoin's algorithm is designed as a guest, not an occupant. The CPU–RAM breathing rhythm — random memory access alternating with serial computation, neither overlapping the other — means neither component is saturated continuously.

The Zero-Cost Miner

A standard home computer has the resources to mine Memcoin without displacing any existing workload. Mining and normal computer use run concurrently without interference.

The mining thread uses CPU capacity that would otherwise sit unused. To someone already using their computer, the perceived marginal cost of one additional mining thread is close to zero.

The participants are people who already own the machines.

The Long View

Gold is finite. It is lost, buried, forgotten — and occasionally found again. The total amount in circulation has never been a clean number. It decreases slowly as rings sink to the ocean floor and coins are sealed in walls. It increases slowly as old mines are reopened and forgotten caches are recovered. The supply is not managed. It simply is.

Memcoin is designed around the same reality. A coin sent to a lost address is not destroyed — it is suspended. The private key exists somewhere. The ledger holds it with perfect fidelity. Unlike physical gold, it cannot be melted down or repurposed. It waits, exactly where it was placed, for a key that may never return.

Every ledger accumulates this damage over time. Keys lost, addresses abandoned, coins frozen. The permanent 7 MEM floor reward slowly returns to circulation what time removes from it — not targeting any specific loss, not attempting to measure the damage. It runs quietly, indefinitely, as the chain's own recovery mechanism.

Gold has survived thousands of years. Its supply is finite, unmanaged, and resistant to debasement. The chain will run for a long time. The parameters will change as the world changes. The commitment written into the first block will not.

On Value

bit is found in memory. It is the result of time and intelligence.

A carbon brain drives a silicon chip. It runs through time, and relies on luck to find the right bit. This is proof that a mind existed.

MEM is for giving. Whatever you want.

---

2. Memory

Before mining begins, a miner chooses what the 64-byte nNonce contains.

A sentence. A line of poetry. A philosophical question. A melody. A record of something that happened. The choice precedes the first hash. The search begins from that starting point and ends when the right bit is found.

A group of people who share a thought agree on the same content. Each signs a name in the final bytes. They search independently, each in a non-overlapping region of the 2^{512} space.

Encoding

64 bytes hold text, music, or both. The byte value alone determines interpretation.

Byte	Meaning
0	Rest (in music) / Terminator (at end)
1–7	Musical notes: C D E F G A B
8–12	Octave: low mid-low middle mid-high high
13–17	Duration: whole half quarter eighth sixteenth
18–22	Dynamics: pp p mf f ff
23	Sharp modifier (♯) — applies to next note

```
24       Flat modifier (♭) — applies to next note
25–29    Tempo: Largo  Andante  Moderato  Allegro  Presto
30       Dotted note — multiplies current duration by 1.5
31       Separator — clears all pending modifiers
32–126   Printable ASCII text
127      Line break
```

State bytes (8–29) persist until changed. Modifiers (23, 24, 30) are consumed on use. Byte 31 takes immediate effect and clears all pending modifiers.

Unset state defaults: middle octave · quarter note · mf · Moderato.

Two extended modes are selected by the first byte (nonce[63]).

Byte 128 marks UTF-8 mode: the remaining 63 bytes encode text in any human script, not only Latin. A 0x00 byte terminates content if present; content may also fill all 63 bytes with no terminator. Drift-corrupted tail bytes decode as U+FFFD rather than failing.

Byte 129 marks packed music mode: the remaining 63 bytes are read as 126 4-bit fields.

```
Nibble  Meaning
──────  ──────────────────────────────────────────
0       Rest
1–7     Notes: C  D  E  F  G  A  B
8       Escape — next nibble is a control selector
9       Inline: middle octave (default-restore)
10      Inline: quarter note (default-restore)
11      Inline: eighth note
12      Inline: half note
13      Inline: high octave
14      Inline: low octave
15      Inline: mf dynamics (default-restore)

After escape (8), selector:
0  EOF             6  Set dynamics  (+1 nibble, 0–4)
1  Sharp ♯         7  Set tempo     (+1 nibble, 0–4)
2  Flat ♭          8  Separator
3  Dotted          9  Line break
4  Set octave  (+1 nibble, 0–4)   10  Natural ♮
5  Set duration (+1 nibble, 0–4)  11  Tie
                                   12–15  Reserved
```

Humans face the universe — vast, without order, without inherent meaning — and answer it. Language names what exists. Music finds pattern in time. Philosophy asks what things are. History records what happened. None of these are properties of the universe. They are what consciousness produces.

A miner faces 2^{512} — a space larger than the count of atoms in the observable universe, disordered, silent. Most of it will never be touched. Most blocks carry random nonces that mean nothing to anyone. Where a mind places a sentence or a melody, meaning exists. Not because the mathematics requires it. Because someone chose it.

The chain records what was chosen. The drift upon the final characters is the physical mark of the work. The time spent belongs only to those who were there.

What was chosen does not belong to time. As long as the chain exists and the language survives, it can be decoded, known, understood, felt — by anyone, at any point forward. The miner may be gone. The choice remains.

---

3. Technical Specification

3.1 Core Parameters

Ticker	MEM
PoW Algorithm	Yesmem
ISA Baseline	x86-64-v3 / ARMv9.0-A
Memory per Thread	8 GiB exactly (2^33 bytes)
Block Time	600 seconds
Max Block Size	32 MB (default), 64 MB (hard cap, ABLA)
Difficulty Algorithm	ASERT (aserti3-2d)
ASERT Half-life	7,200 s (2 hours)
ASERT Anchor	Genesis block (hardcoded)
P2P Port mainnet	9333
RPC Port mainnet	9332
Network Magic	0x4D454D43 (MEMC)
Address Prefix	M (Base58Check, version byte 50)
Genesis nBits	0x20020000
Genesis nTime	1773000000
Codebase	Bitcoin Cash Node v29.0.0

The ASERT difficulty anchor is set to the genesis block. All subsequent difficulty calculations reference genesis height, genesis target, and genesis timestamp.

3.2 Unit System

1 𝕄 = 100 ᴍ = 100,000 ᴋ = 100,000,000 ʙ

Unit	Symbol	Unicode	Definition
MEM	𝕄	U+1D544	base unit
mbit	ᴍ	U+1D0D	1/100 MEM = 1,000,000 ʙ
kbit	ᴋ	U+1D0B	1/100,000 MEM = 1,000 ʙ
bit	ʙ	U+0299	1/100,000,000 MEM

3.3 Proof-of-Work

Block header — 132 bytes

```
hashPrevBlock    32 bytes
hashMerkleRoot   32 bytes
nTime             4 bytes
nNonce           64 bytes
```

nVersion is not present. nBits is not serialized or transmitted; each node computes it from chain state and injects it locally.

Yesmem derives from Yespower 1.0, with a 100-round serial bit-manipulation chain inserted after each random memory access in smix2. The chain's output determines the address of the next memory access, and the chain reloads state from the working set every ten rounds. Removing either breaks the computation.

```
Version                 YESMEM_1_0
N (cost factor)         2,097,152 (2^21)
r (block size)          32
Personalization         Memcoin/Yesmem/v1 (18 bytes, NUL-terminated)
Memory per thread       8 GiB exactly (2^33 bytes)
Mix rounds              100 per call, invoked twice per smix2 iteration
ISA baseline            x86-64-v3 (SSE4.2 + AVX2 + BMI2 + FMA) /
                        ARMv9.0-A (SVE2 + BITPERM)
```

Memory per thread is derived directly from the parameters: 128 × N × r = 128 × 2,097,152 × 32 = 8,589,934,592 bytes = 8 GiB

N is required to be a power of two; 2^21 is the smallest such value whose working set remains within the system RAM of standard PC hardware.

The hash function accepts a 132-byte block header and returns a 256-bit digest:

1. SHA-256(header) → 32-byte digest 2. PBKDF2-SHA-256(digest, pers, 1 iteration) → 128-byte initial B 3. smix(B, r, N, ...): smix1 — sequentially populates the 8 GiB V-table smix2 — ~N/3 iterations, each reading a 4 KiB slot at a pseudorandom index, running blockmix_xor_save, invoking the Memcoin mix chain, then repeating once more within the same iteration 4. HMAC-SHA-256(B_tail, digest) → 256-bit PoW hash

The mix chain

State is three 64-bit words seeded from the current V-slot and address. Each round applies a golden-ratio constant to introduce positional variance, derives two data-dependent masks from the current state, scatters bits through those masks using PEXT and PDEP, then applies a non-linear diffusion step. Every ten rounds, the chain reads two additional words from a data-dependent address across the full 8 GiB table, folding them into state. The final output is derived from all three state words and becomes the next V-table address consumed by smix2.

The chain is strictly serial — each round's output feeds the next round's input. No step can begin before its predecessor completes. SIMD and multi-core acceleration provide no benefit within a single hash.

PEXT and PDEP are native BMI2 instructions on x86. ARM platforms implement the equivalent computation through SVE2 BITPERM. Consensus requires only bit-identical hash output; any implementation on any architecture that produces identical results is a conforming validator.

Every mix call forces ten unpredictable DRAM reads on top of the 4 KiB access performed by blockmix. The next address is never known until the chain completes, so neither prefetch nor pipeline speculation can hide the memory latency.

3.4 CPU–RAM Rhythm

Each outer iteration in smix2 alternates between random memory traffic — a 4 KiB block read from the 8 GiB V-table, plus ten smaller reloads driven by the mix chain — and the serial bit-manipulation chain itself. Neither component idles while the other runs.

At single-thread utilisation, both memory bandwidth and CPU integer throughput run at substantial but non-saturating levels during a hash. RAM duty cycle is materially lower than a pure memory-bound workload, and total thermal output is correspondingly reduced. Long-term mining does not accelerate RAM degradation relative to a typical workstation workload.

3.5 Difficulty and Hashrate

Difficulty is calibrated to one mining thread on commodity PC hardware. Each unit of difficulty corresponds to one committed 8 GiB thread. The difficulty number approximately reflects the number of threads currently mining on the network.

This makes H/s — hashes per second — a precise and fair measure of every participant's contribution. No prefix scaling is required or used. Values are formatted with thousand separators and two decimal places.

3.6 Serial Chain Properties

The 100-round mix chain is strictly serial. Each round's output feeds the next round's input through PEXT/PDEP masks derived from the current state:

```
x(n+1) = f( x(n), v0(n), v1(n) )
```

v0 and v1 evolve only through rotations and the periodic V-table reloads. This data dependency prevents any form of parallelisation across rounds. SIMD instruction sets provide no benefit — and the consensus build explicitly forbids AVX-512, AVX-10, and XOP at compile time to guarantee bit-identical hashes across all validators.

GPUs lack native PEXT and PDEP instructions and must emulate each round through shift-and-mask sequences. GPU throughput is further constrained by VRAM capacity: a single thread requires 8 GiB plus ancillary working set, limiting most consumer cards to at most one concurrent thread.

3.7 Optimisation Ceiling

Three constraints close the optimisation space:

Memory capacity caps thread count at physical RAM ÷ 8 GiB. No amount of computational speed increases this ratio.

The serial mix chain eliminates parallel acceleration paths within a single hash. Each round's output feeds the next through PEXT/PDEP masks derived from state. SIMD, multi-core, and FPGA pipelines all reduce to single-core serial throughput on this chain.

Read-write coupling eliminates prefetch and cache reuse. The chain output determines the next V-table address, so the address is unknown until the chain completes; the prefetch window is zero. The per-mix V-reloads further ensure that even within a single mix call, DRAM reads occur at data-dependent addresses the processor cannot speculate past.

Each constraint reinforces the others. No single constraint can be broken without violating the other two. ASIC designs face the same two absolute physical limits — DRAM random-access latency and the serial round latency of the mix chain — that general-purpose CPUs face. The principal ASIC resistance is the 8 GiB working set: specialised silicon gains nothing from pipelining within a single hash and must pay the same per-thread memory cost.

3.8 Hardware Longevity

General-purpose hardware evolution is constrained by fabrication process and physical law. The performance gap between successive generations is narrow.

Yesmem hashrate is not determined by total memory capacity alone. CPU throughput, memory bandwidth, and memory latency each contribute independently to total output. At single-thread utilisation, memory bandwidth runs below saturation on any commodity platform. Per-thread performance differences between generations are further narrowed at this level.

An improvement in any one dimension is absorbed by the remaining constraints. No single upgrade translates into proportional hashrate gain.

Older hardware is not displaced from hashrate competition by newer generations. As in the broader world of consumer electronics, new products do not obsolete existing machines — hardware leaves circulation gradually, at the end of its natural service life.

This slow generational turnover is reflected in the network's hashrate. Hashrate comes from the computers people already own and use — not single-purpose machines. Mining runs alongside normal computer use, on existing hardware, for as long as the memory remains serviceable.

3.9 Mining Scale Economics

Each mining thread requires 8 GiB of dedicated RAM that cannot be shared across machines. This figure is fixed by the algorithm parameters N and r — independent of CPU speed, memory bandwidth, or hardware generation. Mining capacity scales in strict linear proportion to memory held. Aggregation provides no per-unit memory cost reduction.

Validation overhead follows a variable ratio. Required validation threads scale as:

```
verify_threads = ceil(T × V / S)
```

where T is total mining threads, V is validation time per share (hardware-determined), and S is share interval. V decreases as hardware improves, and the ratio V/S shrinks accordingly. This does not affect the fundamental constraint: mining memory remains 8 GiB per thread regardless of hardware speed.

A 5-minute share interval guarantees at least one share per 10-minute block; a 3-minute interval yields approximately three. The benefit of more frequent share submission is directly paired with higher validation memory cost, which requires higher pool fees and reduces return per share. The advantage and its cost are inseparable.

The benefit of pool aggregation is variance smoothing, and this benefit saturates: a pool finding several blocks per day already provides near-steady income, and further growth adds negligible smoothing. Validation cost, by contrast, grows linearly with pool size without bound. Beyond modest scale, cost continues to accumulate against a benefit that has already plateaued. The domain of rational mining is small pools — not by rule, but by structure.

---

4. Emission Schedule

4.1 Stages 1–3 — Fixed Issuance (Years 0–9)

Stage	Block Range	Reward	Total Issued
1	0 – 157,499	50 MEM	7,875,000 MEM
2	157,500 – 314,999	25 MEM	3,937,500 MEM
3	315,000 – 472,499	12.5 MEM	1,968,750 MEM
Total			13,781,250 MEM

4.2 Stage 4 — Permanent Floor (Block 472,500+)

Fixed at 7 MEM per block, yielding ~367,500 MEM / year in perpetuity.

There is no hard cap. The terminal supply is an equilibrium — the point at which annual issuance equals annual supply lost to irreversible attrition.

$$S_{eq} = 367{,}500 \div 0.007 \approx 52{,}500{,}000 \text{ MEM}$$

This figure is an estimate, not a protocol parameter. The loss rate is an assumption; the real rate is unknown and will vary over time. The equilibrium supply is not enforced by code — it is a consequence of how people use the chain. How much is lost, how much is spent, how much is held: these are human decisions, accumulated over time, that the formula can only approximate.

4.3 Inflation & Loss Rate — MEM vs. Gold (20-Year Intervals)

Assumptions

	MEM	Gold
Annual loss rate	0.70%	0.20%
Annual gross issuance	367,500 (fixed)	~1.65% of stock

Digital asset attrition arises from: private key loss, custodial failures, inheritance gaps, and unrecoverable wallets. Unlike physical commodities, on-chain losses are permanent and unrecoverable.

The 0.70% is estimated between physical cash and bank deposit loss rates, adjusted upward as no centralised recovery mechanism exists.

MEM — Stage 4 Projection

Year	Circulating Supply	Gross Inflation	Loss Rate	Net Inflation
Y9 (Stage 4 start)	13,781,250	2.67%	0.70%	+1.97%
Y29	~18,840,000	1.95%	0.70%	+1.25%
Y49	~23,237,000	1.58%	0.70%	+0.88%
Y69	~27,060,000	1.36%	0.70%	+0.66%
Y89	~30,383,000	1.21%	0.70%	+0.51%
Equilibrium	~52,500,000	0.70%	0.70%	0%

Gold — Reference (stable annual rates)

Gross Inflation	Loss Rate	Net Inflation
~1.65%	~0.20%	~1.45% (perpetual)

Gold maintains a structurally positive net inflation indefinitely. MEM net inflation decelerates asymptotically toward zero as issuance converges with attrition — without any protocol intervention.

4.4 Summary

Property	MEM	Gold
Terminal supply	~52,500,000 (emergent)	No ceiling
Long-run net inflation	→ 0% (emergent)	~1.45% (stable)
Supply floor mechanism	Permanent block reward	Annual mining
Attrition type	Permanent, on-chain	Partial, recoverable

5. Transaction Model

5.1 Address Format

P2PKH only. An address encodes the RIPEMD-160 hash of a compressed public key in Base58Check encoding with version byte 50.

```
Base58:   M...   (version byte 50)
```

P2SH and CashAddr are not supported. The address format is the complete set of valid destinations on this chain.

5.2 Signatures

Both ECDSA and Schnorr signatures are valid. The address format is identical — it encodes the public key hash, not the signature algorithm. ECDSA: 71–72 bytes (DER). Schnorr: 64 bytes (fixed).

5.3 Schnorr and Multi-Party Spending

Through the MuSig protocol, multiple parties can cooperate to produce a single aggregated Schnorr signature. On-chain it is indistinguishable from a standard single-party signature: one address, one 64-byte signature, same transaction size. The number of participants is not visible.

OP_CHECKMULTISIG is not supported. MuSig is the only mechanism for multi-party signing on this chain.

5.4 Time-Locking

Two time-locking mechanisms are in use: OP_CHECKLOCKTIMEVERIFY (CLTV) at the script level, and nLockTime at the transaction level. Their roles are distinct.

CLTV locks TX1 outputs for 432 blocks. nLockTime is used exclusively in TX2, constrained by R8 to within ±30 blocks of the CLTV unlock height.

Setting nLockTime > 0 on an ordinary P2PKH transaction is not permitted. Such a transaction is classified as TX2 and rejected at mempool entry by R7.

5.5 Output Value Floor

Every output on the chain must carry a value of at least 10,000 ʙ (10 kbit). This is a consensus rule: a block containing any output below this floor is invalid and rejected by the entire network.

The rule applies to all transaction types without exception — ordinary P2PKH transfers, TX1 CLTV outputs, TX1 change outputs, and TX2 outputs alike. It is enforced at the output level, not at the transaction level.

The floor bounds the worst-case growth of the UTXO set. Each output that enters the set and is never spent costs at least 10 kbit to create. An attacker attempting to inflate the UTXO set permanently must burn at least 10 kbit per entry, regardless of transaction type or intent.

5.6 Fees

Ordinary P2PKH transactions carry fees. Fee handling is inherited from the Bitcoin Cash Node codebase: the minimum relay fee is a per-node policy parameter, not a consensus rule. The default is 1,000 ʙ (1 kbit) per kilobyte. Consensus requires only that a transaction's outputs do not exceed its inputs; any surplus is claimed by the miner of the including block.

The dust threshold is likewise a per-node policy parameter. Each operator sets it freely, including above the consensus output floor. The 10 kbit floor (5.5) is the absolute lower bound — no policy setting admits an output beneath it.

The instant payment system (TX1 and TX2, Section 6) is exempt from fee policy by protocol. Zero-fee admission applies to those transaction types alone.

6. Zero-Fee Instant Payment

6.1 Philosophy

The payer is the creator and sustainer of network value. The more active the payers, the greater the utility of the chain, the higher the value of MEM, and the greater the long-term income of miners from block rewards alone. Charging payers a fee reduces the incentive to transact.

The instant payment system — TX1 and TX2 — is zero-fee by protocol. A sender who locks funds pays no transaction fee. The recipient who spends those funds pays no transaction fee. Miner income from these transactions is derived entirely from block rewards.

The cheque book draws its design from the logic of a paper cheque. A signed cheque is an irrevocable commitment — the issuer cannot recall it, and the bank does not return funds because a cheque was never presented. The instant payment system encodes the same property at the protocol level.

Once unlocked, a CLTV UTXO is spendable balance. Alice does not redeem it first and spend it second — the unlock is the payment. TX2 is simply the transaction that spends a CLTV UTXO. The two transaction types are separated by protocol: CLTV inputs are zero-fee and cannot be mixed with P2PKH inputs in the same transaction.

6.2 Design

Bob locks the exact payment amount into a single CLTV+P2PKH output for 432 blocks using OP_CLTV. The output is locked to Alice's address. Any remaining balance returns to Bob as P2PKH change in the same transaction. The lock guarantees Bob cannot redirect or replace the funds. Payment settles on-chain when the lock expires.

```
TX1   Bob broadcasts TX1 — TX1 enters the mempool — payment is final
      One CLTV output locked to Alice; optional P2PKH change to Bob
      TX1 confirms on-chain at block H
TX2   The CLTV lock expires at block H+432. Alice broadcasts TX2;
      first possible confirmation is block H+433, at which point
      her UTXO becomes spendable balance. Alice spends it in a
      single zero-fee transaction — combining any unlocked CLTV
      outputs, sending to up to three addresses.
```

Payment is final when TX1 enters the mempool. mempoolreplacement=0 prevents any conflicting transaction from entering the mempool. Bob is not making a promise to pay — the funds are already frozen. Only Alice can spend them.

Traditional payment security looks backward: the deeper a transaction is buried under subsequent blocks, the harder it is to reverse. Memcoin's instant payment looks forward: the CLTV lock makes the future settlement as certain as any confirmed transaction, from the instant TX1 enters the mempool.

TX1 and TX2 form a one-way pipeline:

```
Bob's P2PKH UTXO
    ↓  TX1  (zero-fee; single CLTV output locked to Alice; 432-block CLTV)
CLTV-locked UTXO  (frozen for 432 blocks, locked to Alice)
    ↓  TX2 confirms at H+433 — becomes Alice's spendable balance
TX2: Alice spends to up to three addresses, zero-fee, one block
```

The pipeline closes a complete economic cycle. TX1 takes a single P2PKH UTXO and converts it into a time-locked output assigned to the recipient, with any surplus returned to the sender as change. TX2 takes that UTXO, once unlocked, and returns it to ordinary

P2PKH form — as spendable balance the recipient may use without further restriction.

The constraints are concentrated entirely on the lock side. TX1 enforces R0 through R5b to make the cheque irrevocable and abuse- resistant. TX2 enforces only that its inputs are unlocked CLTV UTXOs (R7) and that its nLockTime is correctly set (R8). It carries no amount limit and no recipient restriction. A single TX2 may combine up to 1000 unlocked CLTV UTXOs from any number of past TX1s and split the total to up to three addresses.

Each TX1 issues exactly one payment. To issue further payments, Bob must create a new TX1.

TX1 entering the mempool is an irrevocable commitment. Bob relinquishes control of the locked output permanently. Alice holds the right to spend it at any time after block H+432. There is no expiry — a UTXO has no timeout mechanism. As long as Alice retains the corresponding private key, the funds remain spendable. If Alice does not spend, the output remains frozen in the UTXO set indefinitely and does not revert to Bob.

This property is stronger than a paper cheque, which typically carries a validity window of several months. A Memcoin cheque does not expire.

6.3 Validation Rules

Rules are divided into two layers. Consensus rules define the structural invariants of TX1 and TX2; a block violating them is invalid and rejected by the entire network regardless of who mines it. Policy rules are per-node parameters that defend against abuse and can be adjusted without a protocol change.

TX1 rules

```
R0   TX1 input count      exactly 1; must be P2PKH             consensus
R1   TX1 fee              == 0                                  consensus
R2   TX1 CLTV locktime    |L − (H+1+432)| ≤ 30 at mempool      consensus
                          |L − (H+432)|   ≤ 30 at pack time
R3   TX1 output[0]        must be CLTV+P2PKH, height-encoded    consensus
R3b  TX1 output count     at most 2; output[1] if present must  consensus
                          be P2PKH back to sender pubkeyhash
R4   TX1 rate limit       ≤ 6 TX1 per sender per 6-block        policy
                          sliding window
R5   TX1 CLTV output value ≤ 1 MEM                              consensus
R5b  TX1 output floor     ≥ mempoolminvalue (default 100 kbit)  policy
```

TX2 rules

```
R6   TX2 fee            == 0                                  consensus
R7   TX2 inputs         all must be CLTV+P2PKH               consensus
R7b  TX2 input count    ≤ 1000                               policy
R8   TX2 nLockTime      within ±30 of each input's CLTV      policy
                        unlock height; locktime range across
                        inputs ≤ 31 blocks
R9   TX2 rate limit     ≤ 10 TX2 per signer per 144-block    policy
                        sliding window
R10  TX2 output count   minimum 1, maximum 3                 consensus
R10b TX2 outputs        all must be P2PKH                    consensus
```

R0: TX1 has exactly one input. A payer whose balance is spread across multiple addresses must consolidate with an ordinary P2PKH transfer before issuing TX1.

R2: Bob sets the CLTV locktime to current_height + 1 + 432 at broadcast time. The ±30 tolerance accommodates propagation delay. This rule is enforced at both mempool admission and block validation. A TX1 that remains in the mempool long enough for the chain to advance beyond the pack window must be reconstructed with an updated locktime.

R3: The CLTV+P2PKH output uses a height-encoded locktime only. Time-encoded locktimes are rejected at template match.

R3b: If Bob's input exceeds the payment amount, output[1] returns the difference to Bob's own address. No other output structure is permitted. One TX1 issues exactly one payment to exactly one recipient.

R4: Rate limit is checked against a sliding window of the 6 most recent blocks plus the mempool, attributed to the sender address. At most 6 TX1s per sender address per window.

R5: The CLTV output value ceiling of 1 MEM is enforced at both mempool admission and block validation. A TX1 whose CLTV output exceeds 1 MEM is rejected at mempool entry and, if packed into a block, causes that block to be rejected by the entire network.

R5b: mempoolminvalue sets the operational floor for TX1 acceptance at each node. The default is 100 kbit. Any TX1 whose CLTV output falls below the local threshold is rejected at mempool entry. The global consensus floor (see 5.5) applies beneath this: mempoolminvalue cannot be configured below 10 kbit.

R7: TX2 cannot mix CLTV and P2PKH inputs. The CLTV lifecycle is strictly one-directional: P2PKH → CLTV+P2PKH → P2PKH. CLTV outputs never chain or nest.

R8: When TX2 combines multiple CLTV inputs, their locktime heights must span no more than 31 blocks. Inputs whose heights differ by more than 31 cannot share a valid nLockTime and must be spent in separate TX2s. In practice, TX1 transactions issued within the same ~5-hour period can be merged; those issued further apart must be spent separately. A reorganisation deep enough to invalidate TX2 is highly unlikely on an active chain — if it occurs, Bob re-broadcasts TX1 and Alice waits for re-confirmation before broadcasting TX2.

R9: Rate limit is checked against a sliding window of the 144 most recent blocks plus the mempool, attributed to the signing address.

6.4 Network Governance

mempoolminvalue serves two independent functions.

The first is operational: each node operator sets a floor for TX1 acceptance. Any TX1 whose CLTV output value falls below the local threshold is rejected at mempool entry — it does not propagate, does not reach miners, and does not consume network resources. This is the primary defence against zero-fee spam attacks on the instant payment system. Each operator sets this value independently. No one else determines it.

The second is optional: when the community disagrees on any question — a protocol parameter, an upgrade, or anything else — mempoolminvalue can serve as an expression platform. The question can be anything. A technical upgrade. A community position. Whether pineapple belongs on pizza.

Because Yesmem requires only standard PC hardware, miners and users are the same population. A node operator's configuration is also a miner's configuration. The vote and its execution are the same action.

The mechanism:

1. A proposer publishes a question, a signal value, and a start block. The signal value sits above the current network default and below the proposed threshold — for example, 150 kbit if the current default is 100 kbit and the proposed threshold is 200 kbit.

2. Those who support the position set their local mempoolminvalue to the proposed threshold before the start block.

3. The proposer broadcasts one signal TX1 per block for 144 blocks from the start height, using 144 independent UTXOs prepared in advance.

4. For each signal TX1, the proposer records whether the immediately following block includes it or not. A block that does not include it is counted as a rejection. The transaction remains in non-supporters' mempools and will eventually confirm in a later block — the count measures each block's mining policy, not final settlement.

```
Rejected ≥ 96 (2/3)   position adopted
Rejected < 96          position not adopted
```

The window matches the TX2 rate-limit window and provides sufficient statistical margin.

An example. A community question: is pineapple acceptable on pizza?

Proposer announces: signal value 150 kbit, start block #1000 Supporters set mempoolminvalue = 200 kbit before block #1000 Those who disagree make no change

For 144 blocks, the proposer broadcasts one 150 kbit TX1 per block. A supporter's block does not include it — the transaction never entered their mempool. A non-supporter's block includes it normally. Transactions not included by their corresponding block remain in non-supporters' mempools and confirm in subsequent blocks.

98 of 144 signal transactions were not included by their corresponding block. 98 ≥ 96. Position adopted.

This result carries no enforcement power. It cannot compel any action. It records that during those 144 blocks, more than two thirds of the network's hashrate set their local configuration to reflect this position.

Three properties of the mechanism:

```
Objective       any person who counts independently arrives
                at the same number
Not executable  no code triggers automatically, no committee
                interprets the outcome
Inviolable      the record, once written, cannot be altered
```

A position held by two thirds of the network's mining power is detectable across 144 blocks.

The governance window closes after the 144-block observation period. The result is determined by block inclusion during that window, not by final settlement of the signal transactions. No other mechanism is required or exists.

6.5 Security Model

```
Bob double-spends TX1           OP_CLTV — block invalid, network rejects
Bob double-spends TX1 input     block reward exceeds attack gain
(before TX1 confirms)           economically irrational at any tx size
Miner packs illegal TX          block rejected, miner loses block reward
Chain reorganisation            R8 ±30 tolerance — Alice re-signs,
                                no funds at risk
UTXO bloat (any type)           consensus floor 10 kbit per
                                output; cost = 10 kbit per permanent UTXO
TX1 spam (rate)                 R4 — 6 TX1 per sender per 6-block window
TX1 spam (value)                R5b — mempoolminvalue floor (default
                                100 kbit)
TX1 multiple inputs             R0 — exactly one input enforced
Non-CLTV TX2 input              R7 — all inputs must be CLTV+P2PKH
TX2 mixed inputs                R7 — CLTV lifecycle is one-directional
Alice does not spend output     funds remain frozen in UTXO set
                                permanently; Bob has no path to reclaim
```

The primary protection against double-spend is OP_CLTV, a consensus rule: any block containing a transaction that spends a CLTV-locked output before its unlock height is invalid and rejected by the entire network. mempoolreplacement=0 is a protocol default that prevents conflicting TX1 transactions from entering the mempool — it reinforces but does not replace the consensus guarantee.

The locktime tolerance (R2) and output value ceiling (R5) are likewise enforced at both mempool admission and block validation. A TX1 that violates either is rejected at the node and, if packed into a block, causes that block to be rejected by the entire network.

The abandoned-output attack is bounded at the consensus layer. An attacker who locks funds into TX1s directed at addresses they control and never broadcasts TX2 permanently inflates the UTXO set. The output value floor prices this attack across all transaction types: each 10 kbit spent creates at most one permanent UTXO. The mempoolminvalue floor (R5b, default 100 kbit) raises the operational cost by a further factor of ten for TX1 transactions.

Network-level spam is addressed through protocol rate limits and miner social consensus. R4 caps TX1 throughput per sender at 6 per 6-block window. R9 caps TX2 throughput per signer at 10 per 144-block window. Yesmem prevents large mining pools — miners are distributed individuals who can raise mempoolminvalue in response to an attack without a hard fork. Legitimate users are not affected because the threshold is publicly known.

6.6 Zero-Confirmation Security

A standard P2PKH transaction entering the mempool carries moderate zero-confirmation risk. Validity is established at mempool entry — signature correct, UTXO exists, fee sufficient — but the sender retains the ability to broadcast a conflicting transaction spending the same UTXO. mempoolreplacement=0 raises the cost of this attack but does not eliminate it at the protocol level.

TX1 is different in kind, not degree. When TX1 enters the mempool, Bob's funds are frozen in a CLTV-locked output assigned to Alice. mempoolreplacement=0 prevents any conflicting transaction from entering the mempool. A miner colluding with Bob could bypass the mempool and build a block that double-spends Bob's input — but at any realistic cheque value, this yields less than the foregone block reward.

Alice does not need to wait for TX1 to confirm on-chain for everyday small transactions. At this scale, the maximum gain from a miner-coordinated double-spend is negligible compared to the block reward — no rational miner will sacrifice a block reward to capture a fraction of it. Only Alice can spend the locked output. In a single zero-fee transaction, she may combine any unlocked CLTV outputs, split the total to up to three addresses. Confirmed in one block.

	P2PKH	TX1 (instant payment)
Mempool entry proves	sig + UTXO	sig + UTXO + CLTV lock
Sender can double-spend	possible	economically irrational — block reward exceeds max gain
Zero-conf safety	moderate	sufficient for everyday use

For larger amounts, recipients naturally wait for one confirmation before releasing goods or services. The design aligns incentives such that zero-confirmation is safe precisely where it is used, and confirmation is sought precisely where the value justifies it.

6.7 Complete Flow

```
TX1 broadcast    TX1 enters mempool — Bob loses control permanently
Block H          TX1 confirmed on-chain
H+1 to H+432     CLTV output frozen — Alice cannot spend yet
Block H+432      CLTV lock expires — Alice broadcasts TX2
Block H+433      TX2 confirms
```

Any unspent output remains spendable by Alice at any time after H+432. Bob has no path to reclaim it.

7. UTXO Lifecycle

7.1 Tiers

The temperature system assigns each output a storage tier proportional to its age and value.

```
Normal → Cold → Ice
```

The path is forward-only.

```
Normal   In-memory chainstate. Every confirmed output enters here.

Cold     Compressed address-level index on disk. Outputs at the same
         address share a single record. Spending requires no block read.

Ice      Block heights only. Spending triggers a block read to
         reconstruct the output data on demand.
```

The demotion schedule is a deterministic function of value and confirmation height. Every node computes it identically.

7.2 Temperature

```
COOLING_RATE = 2000 в per block
```

```
temperature(h) = value − COOLING_RATE × (h − heightAnchor)
```

For P2PKH outputs, heightAnchor is the confirmation height. For CLTV+P2PKH outputs, heightAnchor is the unlockHeight — a locked output begins to cool only after it becomes spendable.

A UTXO is demoted from Normal at:

```
demotionBlock = heightAnchor + floor(value / COOLING_RATE) + 1
```

Normal tier lifetimes:

```
10 kbit (minimum)        5 blocks   ≈ 50 minutes
100 kbit                50 blocks   ≈ 8 hours
1 mbit                 500 blocks   ≈ 3.5 days
1 𝕄             50,000 blocks   ≈ 347 days
```

On entry to the Cold tier:

```
iceDemotionBlock = coldBlock + protectionPeriod + floor(value / COOLING_RATE) +
1
```

```
protectionPeriod   Default: 12,960 blocks (~90 days). Operator-adjustable.
```

Cold tier lifetime for a 1 𝕄 output:

```
Normal cooling      50,000 blocks   ≈ 347 days
Cold protection     12,960 blocks   ≈  90 days
Cold cooling        50,000 blocks   ≈ 347 days
Total to Ice                        ≈ 784 days
```

7.3 Spending

A Cold UTXO requires a disk read. An Ice UTXO requires reading the blocks listed in the ice index to reconstruct the output data on demand.

7.4 utxofall

A node that cannot reconstruct an Ice UTXO locally requests the data from the node that propagated the spending transaction. That node already validated the transaction before relaying it. The exchange is point-to-point and not broadcast to the network.

```
UTXO_REQUEST     txid + vout
UTXO_RESPONSE    txid + vout + value + scriptPubKey
```

Whether a node retains full block history, has pruned, or has disabled the ice index is not visible to the network. All nodes mine, validate transactions, and relay blocks on equal terms.

7.5 Reorg Handling

Reorg data is retained for 432 blocks — the same value as the TX1 lock period.

7.6 Consensus and Configuration

The temperature formula, COOLING_RATE, and the Normal-to-Cold demotion schedule are consensus rules. Every node applies them identically.

The protection period and the Ice index depth are configuration parameters that operators may adjust without a protocol change.

8. Network Security

8.1 51% Attack Cost

The cost of a 51% attack scales linearly with network size. Each attacking thread requires 8 GiB of dedicated RAM — memory cannot be shared between threads. Block validation carries the same memory requirement as mining. Attack cost compounds: mining capacity plus verification capacity, neither reducible.

Beyond a sufficient network size, the attack transitions from expensive to physically unavailable. Large-memory cloud capacity is finite and largely committed. The 8 GiB per-thread floor excludes the majority of consumer botnets, whose typical members cannot satisfy the dedicated memory requirement.

9. Protocol and Implementation

Memcoin defines a protocol, not a software distribution. Anyone may write and distribute software that implements the protocol — wallets, payment interfaces, payment tools, block explorers, or mining clients. Conformance is determined by the protocol rules alone.

Upgrades follow the same principle. A change to the protocol is adopted when miners include it in blocks and users accept those blocks. The network decides by running the software it chooses to run.

Memcoin is a new ledger. It shares the philosophy of its predecessors — peer to peer, permissionless, no trusted third party — and carries nothing else forward. The chain starts at block zero.

— Memcoin (MEM) · 2026